

quantizedSAM

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1 Quantized SAM

1.1 Sharpness-Aware Minimization (SAM)

Let $S = \{(x_i, y_i)\}_{i=1}^n$ be the training set and $\mathcal{L}_S(w)$ the empirical loss of a model with parameters w . Sharpness-Aware Minimization (SAM) seeks parameters that achieve low loss *and* reside in flat regions of the loss landscape by minimizing the worst-case loss within a small neighborhood of w :

$$\min_w \mathcal{L}_S^{\text{SAM}}(w) + \lambda \|w\|_2^2, \quad \mathcal{L}_S^{\text{SAM}}(w) := \max_{\|\epsilon\|_p \leq \rho} \mathcal{L}_S(w + \epsilon), \quad (1)$$

where $\rho > 0$ controls the neighborhood radius and $p \in [1, \infty]$ specifies its geometry. The inner maximization is commonly approximated by a first-order expansion around w :

$$\mathcal{L}_S(w + \epsilon) \approx \mathcal{L}_S(w) + \epsilon^\top g, \quad g := \nabla_w \mathcal{L}_S(w), \quad (2)$$

which yields the dual-norm solution

$$\epsilon^*(w) \approx \arg \max_{\|\epsilon\|_p \leq \rho} \epsilon^\top g. \quad (3)$$

For the common choices $p = 2$ and $p = \infty$, the maximizer reduces to

$$\epsilon^*(w) = \rho \frac{g}{\|g\|_2} \quad (p = 2), \quad \epsilon^*(w) = \rho \text{sign}(g) \quad (p = \infty). \quad (4)$$

SAM then updates w using the gradient evaluated at the perturbed point $w + \epsilon^*(w)$. Following the standard SAM approximation, the dependence of $\epsilon^*(w)$ on w is ignored when taking the outer gradient, resulting in a practical two-pass procedure: (i) compute $g = \nabla_w \mathcal{L}_S(w)$ and form $\epsilon^*(w)$; (ii) compute $\nabla_w \mathcal{L}_S(w + \epsilon^*(w))$ and apply the optimizer update.

1.2 Quantization and Why Naïvely Combining SAM Can Fail

We consider weight quantization via a (uniform affine) quantizer $Q(\cdot)$, e.g.,

$$Q(w) = s \cdot \text{clamp}(\text{round}(w/s) + z, q_{\min}, q_{\max}), \quad (5)$$

where s is the scale (step size), z the zero-point, and $\text{round}(\cdot)$ induces a piecewise-constant mapping. Let $q := Q(w)$ denote the quantized weights. In deployment, the effective model is governed by q rather than w . In quantization-aware training (QAT), $Q(\cdot)$ is inserted in the forward pass (fake quantization), while gradients are backpropagated through Q using the straight-through estimator (STE), typically approximating $\partial Q(w)/\partial w \approx I$ (optionally with a clipping mask).

Naïve SAM with quantization. A straightforward attempt to “apply SAM under quantization” is to perturb w but evaluate the loss after quantization, e.g.,

$$\min_w \max_{\|\epsilon\|_p \leq \rho} \mathcal{L}_S(Q(w + \epsilon)), \quad (6)$$

or to compute the SAM perturbation using gradients obtained under QAT/STE and then quantize the perturbed weights. However, due to the discretization induced by rounding, such naïve combinations can suffer from two major failure modes.

(1) Perturbation diminishment (plateaus due to discretization). Because $Q(\cdot)$ is piecewise constant, small perturbations often do not change the quantized weights. For uniform quantization with step size Δ (possibly per-channel), if a coordinate w_i lies away from bin boundaries, then for any perturbation satisfying $|\epsilon_i| < \Delta_i/2$ we have

$$Q(w_i + \epsilon_i) = Q(w_i), \quad \Rightarrow \quad \mathcal{L}_S(Q(w + \epsilon)) = \mathcal{L}_S(Q(w)). \quad (7)$$

Consequently, the inner maximization in (6) becomes flat over a nontrivial region, and the SAM perturbation may be effectively “absorbed” by the quantization error, yielding negligible change in the evaluated loss.

(2) Perturbation mismatch (pre-quant perturbation vs post-quant objective). Even when ϵ is computed using gradients under QAT, the direction of perturbation can mismatch the true worst-case direction for the quantized objective. In particular, the inner maximization in (6) concerns the sensitivity of $\mathcal{L}_S(Q(\cdot))$, whereas the perturbation is typically constructed in the continuous w -space using an STE surrogate. Moreover, clamping/saturation in Q can distort the local sensitivity, causing gradients used to build ϵ to be inconsistent with how $Q(w + \epsilon)$ actually changes (or fails to change) after rounding.

Implication. These issues motivate defining sharpness-aware objectives *directly* in the quantized-weight domain (e.g., perturbing $q = Q(w)$ rather than w) or introducing lattice-/rounding-aware adversaries that ensure the inner maximization produces meaningful changes under discretization. Therefore, to obtain a meaningful sharpness-aware objective under discretization, we redefine the SAM neighborhood *after* quantization by perturbing the quantized weights $q = Q(w)$ (rather than the full-precision weights w), which avoids perturbation diminishment and better matches the deployed model.

1.3 Quantization-Geometry Sharpness-Aware Minimization

Setup and notation. Let $S = \{(x_i, y_i)\}_{i=1}^n$ be the training set and $\mathcal{L}_S(\cdot)$ the empirical loss. We denote by w the full-precision weights and by $q(w) := Q(w)$ their quantized counterpart produced by a (weight) quantizer $Q(\cdot)$. We assume a groupwise quantization structure (e.g., per-channel or per-layer), partitioning parameters into G disjoint groups $\{w_g\}_{g=1}^G$ and applying (possibly different) quantizers $Q_g(\cdot)$:

$$q_g(w) := Q_g(w_g), \quad q(w) = (q_1(w), \dots, q_G(w)). \quad (8)$$

Let $\Delta_g > 0$ denote the effective quantization step (scale) for group g .¹ We write $D := \text{blockdiag}(\Delta_1 I, \dots, \Delta_G I)$.

Motivation: beyond post-quant perturbations (Case 3). Post-quant perturbation methods construct the SAM neighborhood directly in the quantized domain as $\max_{\|\delta\|_p \leq \rho_q} \mathcal{L}_S(q + \delta)$. While this avoids perturbation diminishment due to discretization, it treats all groups isotropically and does not explicitly reflect the heterogeneous quantization granularity across groups (e.g., different Δ_g). To address this, we define the adversarial neighborhood in an *index/level* parameterization and map it to the quantized-weight space via the groupwise steps.

Objective: quantization-geometry qSAM. We define the following sharpness-aware objective under quantization:

$$\min_w \mathcal{L}_S^{\text{QG}}(w) + \lambda \|w\|_2^2, \quad \mathcal{L}_S^{\text{QG}}(w) := \max_{\forall g, \|u_g\|_p \leq \rho_g} \mathcal{L}_S(q(w) + Du), \quad (9)$$

where $u = (u_1, \dots, u_G)$ is an *index-space* perturbation, $\rho_g > 0$ is a groupwise radius, and $\delta := Du$ is the induced perturbation in the quantized-weight space. Intuitively, $\|u_g\|_p \leq \rho_g$ constrains the perturbation in units of quantization levels, while D converts it to the corresponding perturbation magnitude in the quantized weights.

Inner maximization via first-order approximation. Let $q := q(w)$ and define the gradient with respect to quantized weights

$$g_q := \nabla_q \mathcal{L}_S(q) = \frac{1}{|S|} \sum_{(x,y) \in S} \nabla_q \ell(f(x; q), y). \quad (10)$$

Using a first-order expansion around q ,

$$\mathcal{L}_S(q + Du) \approx \mathcal{L}_S(q) + u^\top D^\top g_q = \mathcal{L}_S(q) + \sum_{g=1}^G u_g^\top (\Delta_g g_{q,g}), \quad (11)$$

¹For uniform (affine) quantization, Δ_g corresponds to the scale s_g , possibly after accounting for symmetric/asymmetric conventions.

where $g_{q,g}$ denotes the block of g_q corresponding to group g . Thus the inner maximization in (9) decomposes across groups:

$$u_g^*(w) \approx \arg \max_{\|u_g\|_p \leq \rho_g} u_g^\top (\Delta_g g_{q,g}), \quad g = 1, \dots, G. \quad (12)$$

Let r be the Hölder conjugate of p ($\frac{1}{p} + \frac{1}{r} = 1$). For $1 < p < \infty$, the dual-norm solution yields

$$u_g^*(w) = \rho_g \cdot \frac{\text{sign}(g_{q,g}) \odot |g_{q,g}|^{r-1}}{\|g_{q,g}\|_r^{r-1}}, \quad \delta_g^*(w) := \Delta_g u_g^*(w). \quad (13)$$

In particular, the induced perturbation in the quantized-weight domain becomes

$$\delta_g^*(w) = \rho_g \Delta_g \cdot \frac{\text{sign}(g_{q,g}) \odot |g_{q,g}|^{r-1}}{\|g_{q,g}\|_r^{r-1}}. \quad (14)$$

The common special cases reduce to

$$\delta_g^*(w) = \rho_g \Delta_g \frac{g_{q,g}}{\|g_{q,g}\|_2} \quad (p = 2), \quad \delta_g^*(w) = \rho_g \Delta_g \text{sign}(g_{q,g}) \quad (p = \infty). \quad (15)$$

Compared to isotropic post-quant perturbations, (14) yields a distinct form that (i) normalizes gradients *within each quantization group* and (ii) scales perturbations by the groupwise quantization step Δ_g , thereby explicitly reflecting heterogeneous quantization granularity.

Two-pass update with SAM/STE approximations. Let $q^+(w) := q(w) + \delta^*(w)$ with $\delta^*(w) = (\delta_1^*(w), \dots, \delta_G^*(w))$. We implement (9) using a SAM-style two-pass update. First, we compute g_q at $q = Q(w)$ and construct $\delta^*(w)$ using (14)–(15). Second, we evaluate the sharpness-aware gradient at the perturbed quantized weights

$$g_{q^+} := \nabla_{q^+} \mathcal{L}_S(q^+) = \frac{1}{|S|} \sum_{(x,y) \in S} \nabla_{q^+} \ell(f(x; q^+), y). \quad (16)$$

For the outer update, the exact chain rule yields

$$\nabla_w \mathcal{L}_S(q^+(w)) = (J_Q(w) + J_{\delta^*}(w))^\top g_{q^+}, \quad (17)$$

where $J_Q(w) := \frac{\partial Q(w)}{\partial w}$ and $J_{\delta^*}(w) := \frac{\partial \delta^*(w)}{\partial w}$. Following the SAM approximation, we ignore the dependence of $\delta^*(w)$ on w and drop $J_{\delta^*}(w)$:

$$\nabla_w \mathcal{L}_S(q^+(w)) \approx J_Q(w)^\top g_{q^+}. \quad (18)$$

In QAT, we backpropagate through $Q(\cdot)$ using the straight-through estimator (STE), approximating $J_Q(w)$ as the identity (optionally with a clipping mask $m(w) \in \{0, 1\}^d$). Thus,

$$J_Q(w)^\top g_{q^+} \approx m(w) \odot g_{q^+}, \quad (19)$$

and the final update with weight decay is

$$w \leftarrow w - \eta \left(m(w) \odot g_{q^+} + 2\lambda w \right). \quad (20)$$

Algorithm.

Algorithm 1: Quantization-Geometry qSAM (QG-qSAM)

Input: weights w , quantizer Q , group steps $\{\Delta_g\}_{g=1}^G$, radii $\{\rho_g\}_{g=1}^G$, norm p , learning rate η , weight decay λ .

Repeat:

1. **Quantize:** $q \leftarrow Q(w)$.
 2. **Pass 1:** compute $g_q \leftarrow \nabla_q \mathcal{L}_S(q)$.
 3. **Construct perturbation:** for each group g , compute δ_g^* using (14) (or (15)) and set $\delta^* \leftarrow (\delta_1^*, \dots, \delta_G^*)$.
 4. **Perturb:** $q^+ \leftarrow q + \delta^*$.
 5. **Pass 2:** compute $g_{q^+} \leftarrow \nabla_{q^+} \mathcal{L}_S(q^+)$.
 6. **Outer update (STE):** $w \leftarrow w - \eta(m(w) \odot g_{q^+} + 2\lambda w)$.
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1.4 Rounding-Aware (Lattice) Extension

From continuous index perturbations to integer level shifts. In Equation (9)–Equation (15), we introduced an *index-space* perturbation u_g and used a continuous relaxation $u_g \in \mathbb{R}^{d_g}$ to obtain the closed-form inner maximizer via dual norms. However, under uniform quantization the quantized weights take values on a lattice with step Δ_g , and meaningful rounding-induced changes correspond to *integer* level shifts. We therefore consider a rounding-aware variant that enforces $u_g \in \mathbb{Z}^{d_g}$, so that the induced perturbation $\delta_g = \Delta_g u_g$ corresponds to valid lattice moves in the quantized-weight domain.

Rounding-aware objective. We define the rounding-aware sharpness-aware objective as

$$\min_w \mathcal{L}_S^{\text{RA}}(w) + \lambda \|w\|_2^2, \quad \mathcal{L}_S^{\text{RA}}(w) := \max_{\forall g, u_g \in \mathbb{Z}^{d_g}, \|u_g\|_p \leq \kappa_g} \mathcal{L}_S(q(w) + Du), \quad (21)$$

where $q(w) = Q(w)$, $D = \text{blockdiag}(\Delta_1 I, \dots, \Delta_G I)$, and κ_g is an integer budget controlling the magnitude of allowable level shifts in group g . Compared to (9), (21) replaces the continuous neighborhood by a lattice-constrained one, ensuring the inner maximization produces *nontrivial* rounding-consistent perturbations.

First-order surrogate and lattice inner maximization. Let $q := q(w)$ and $g_q := \nabla_q \mathcal{L}_S(q)$ as in (10). A first-order approximation gives

$$\mathcal{L}_S(q + Du) \approx \mathcal{L}_S(q) + u^\top D^\top g_q = \mathcal{L}_S(q) + \sum_{g=1}^G u_g^\top (\Delta_g g_{q,g}). \quad (22)$$

Hence the surrogate inner problem decomposes across groups:

$$u_g^*(w) \approx \arg \max_{u_g \in \mathbb{Z}^{d_g}, \|u_g\|_p \leq \kappa_g} u_g^\top (\Delta_g g_{q,g}), \quad g = 1, \dots, G. \quad (23)$$

Since (23) is an integer optimization, we adopt tractable constructions tailored to common norms.

Case $p = \infty$: sign level-shifts (coordinatewise bounded). When $p = \infty$, the constraint $\|u_g\|_\infty \leq \kappa_g$ implies each coordinate can shift by at most κ_g quantization levels. The maximizer of the linear surrogate in (23) is obtained coordinatewise:

$$u_{g,i}^* = \kappa_g \operatorname{sign}(g_{q,g,i}), \quad \delta_{g,i}^* = \Delta_g u_{g,i}^* = \kappa_g \Delta_g \operatorname{sign}(g_{q,g,i}). \quad (24)$$

Thus, $p = \infty$ yields rounding-aware perturbations that perform *integer* sign shifts of up to κ_g levels per coordinate.

Case $p = 2$: top- k signed level-shifts (energy bounded). When $p = 2$, the constraint $\|u_g\|_2 \leq \kappa_g$ limits the ℓ_2 energy of integer level shifts. The exact integer maximizer is in general nontrivial; we therefore use a simple and effective approximation that (i) selects a subset of coordinates with the largest $|g_{q,g,i}|$ and (ii) applies one-level signed shifts to them. Specifically, let $\mathcal{I}_g$ be the set of indices corresponding to the top- K_g values of $|g_{q,g,i}|$, where

$$K_g := \min\{d_g, \kappa_g^2\}, \quad (25)$$

so that the constructed integer vector satisfies $\|u_g\|_2 = \sqrt{K_g} \leq \kappa_g$. We then define

$$u_{g,i}^* = \begin{cases} \operatorname{sign}(g_{q,g,i}), & i \in \mathcal{I}_g, \\ 0, & \text{otherwise,} \end{cases} \quad \delta_g^* = \Delta_g u_g^*. \quad (26)$$

This top- k construction is a rounding-consistent analogue of the continuous $p = 2$ solution: it concentrates the limited integer “energy” on the most sensitive coordinates while guaranteeing that the perturbation corresponds to valid quantization-level moves.

Two-pass update (as in QG-qSAM). Given $\delta^*(w) = (\delta_1^*(w), \dots, \delta_G^*(w))$ constructed by (24) (for $p = \infty$) or (26) (for $p = 2$), we form $q^+(w) := q(w) + \delta^*(w)$ and compute

$$g_{q^+} := \nabla_{q^+} \mathcal{L}_S(q^+) = \frac{1}{|S|} \sum_{(x,y) \in S} \nabla_{q^+} \ell(f(x; q^+), y). \quad (27)$$

The outer update follows the same SAM/STE approximations as Equations (18) and (19), yielding

$$w \leftarrow w - \eta \left(m(w) \odot g_{q^+} + 2\lambda w \right). \quad (28)$$

Algorithm.

Algorithm 2: Rounding-Aware qSAM (RA-qSAM)

Input: weights w , quantizer Q , group steps $\{\Delta_g\}$, integer budgets $\{\kappa_g\}$, norm $p \in \{2, \infty\}$, learning rate η , weight decay λ .

Repeat:

1. **Quantize:** $q \leftarrow Q(w)$.
 2. **Pass 1:** compute $g_q \leftarrow \nabla_q \mathcal{L}_S(q)$.
 3. **Construct integer level-shift perturbation:**
 - if $p = \infty$: set $\delta_g^* \leftarrow \kappa_g \Delta_g \text{sign}(g_{q,g})$ for all g (24);
 - if $p = 2$: choose $\mathcal{I}_g$ as top- K_g indices of $|g_{q,g}|$ with $K_g = \min\{d_g, \kappa_g^2\}$ and set $\delta_g^* \leftarrow \Delta_g u_g^*$ by (26).
 4. **Perturb:** $q^+ \leftarrow q + \delta^*$.
 5. **Pass 2:** compute $g_{q^+} \leftarrow \nabla_{q^+} \mathcal{L}_S(q^+)$.
 6. **Outer update (STE):** $w \leftarrow w - \eta(m(w) \odot g_{q^+} + 2\lambda w)$.
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